

Supplementary material to Grain-scale kinematics, ice-bond failure and damage evolution in ice-bonded granular media under in-situ uniaxial compression

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The analysis code and the tables reproduced here are at <https://github.com/AmiraliKhosrozadeh/ice-bonded-granular-microCT>. Specimen labels, load-step numbering and the axial convention (punch at the top) are those of the paper.

S1. Method settings and provenance

Table S1 lists, for each analysis of the paper, what it takes in, what it returns, the software that performs it and the settings that fix the result. Every entry not attributed to a package is the authors' own code.

Table S1: Inputs, outputs, software and settings of each analysis.

Analysis	Input	Output	Software and settings
Phase classification	filtered greyscale volume	phase map (air, ice, grain)	own code; 3 ³ median; valley thresholds per scan for beads, three-class Otsu for sand
Bead labels	phase map	grain labels	own code; marker-controlled watershed on the distance transform; shape gate on fill, radius and roughness

Analysis	Input		Output		Software and settings
Sand labels	greyscale volume		grain labels		own 3D U-Net, border-core, trained on synthetic packings; selected against watershed, a uniform-intensity U-Net, ParticleSeg3D and Cellpose-SAM (Table 2 of the paper)
Bead DDIC	greyscale N , $N+1$; of N	scans labels	per-grain translation used	Φ ;	SPAM 0.9 <code>spam-ddic</code> ; full affine, label dilation 2, margin 15 voxels, 300 iterations, two-level multiscale, updated gradient, per-grain seed; converged at return status 2, usable at 1; grey scales mapped on a copy where they differed
Sand tracking	labels scans; lengths	of both column	matched pairs	grain	own code; affine predictor from column length, refinement over the 12 nearest accepted matches, half-spacing tolerance, volume ratio 0.6–1.6, reciprocity required
Strain and rotation	accepted translations		$\mathbf{G}$, $\mathbf{E}$, E_{eq} , rotation		own code; least squares with intercept over 12 (beads) or 16 (sand) nearest neighbours; Green–Lagrange; Eq. (7) of the paper; composed $1 \rightarrow N$ fields re-fitted at scan-1 positions; rigid column motion removed
Bond census	phase map, labels, matched pairs	map, la- bels, matched pairs	throats, $\bar{z}$, neck radius, sur- vival, coverage pair, class		own code; lens footprint $0.40R$; neck on the plane through the narrowest free point; survival classes at 12 voxels; coverage in a 1–3 voxel shell on a $3\times$ upsampled grid; $C_{\text{HI}} \in \{0.5, 0.7, 0.8, 0.9\}$, reference 0.8; answerable at gap ≥ 7 voxels
Breakage clustering	broken-bond midpoints		cluster RMS, p	size,	scikit-learn DBSCAN, min. samples 5, clusters ≥ 6 ; neighbourhood one grain diameter, sweep 0.6–2.0; 400 resamplings of the bonds present; Benjamini–Hochberg across specimens

Analysis	Input	Output	Software and settings
Core structure	phase map	ε_c , ϕ_{ice} , ρ , S_V , spanning air, k/k_0	own code; core radius 0.85 of the 98th-percentile grain-cloud radius, height band 0.20–0.80 of the column, re-centred per scan; S_V by voxel-face count; 26-connectivity on the 2×-binned map, spanning between the terminal 5 % bands; Carman form with $k_K = 5$
Crack classes	phase map, la- bels	void, surface and body crack, separation	own code; bead envelope (closing $1.25R$, holes filled, 4-voxel skin, one bead diameter clear of each platen); void ceiling 2 mm ³ ; crack width $\leq$ 0.25 mm; skin 20 voxels; bodies < 100 voxels dropped; change from the unloaded scan reported
Ice tortuosity	ice phase of inte- rior cubes	$D_{eff,z}$, τ_{ice}	pumapy 3.2.2 <code>compute_continuum_tortuosity</code> ; 2×-binned map; 128 ³ voxels (6.3 mm); three cubes at 0.28–0.72 of the column on the axis, $\geq 98\%$ inside the specimen; $D_0 = 1$ in ice, 0 elsewhere; $c = 0/1$ on the axial faces, periodic sides; conjugate gradient, tolerance 10^{-6} , 2×10^5 iterations; accepted if $D_{eff,z} > 0$ and $\tau \geq 1$; mean over accepted cubes

Analysis	Input	Output	Software and settings
Ice-path map	phase map	geodesic tortuosity of the ice relative to the unloaded column	own code, scikit-image MCP_Geometric; grain envelope (per-slice closing $1.25R$, holes filled, 2-voxel skin; sand pulled in 1 mm), largest 26-connected ice body; cost $1/(\text{ice fraction in a } 0.5 \text{ mm window, normalised by the window inside the envelope, floor } 0.1)$; seeds on the first ice voxel of every axial line within 3 mm of the support; ratio of path to axial distance from the seed surface, divided by the per-height median of the unloaded scan; drawn where above the unloaded scan's 99.5th percentile at the same height (1 mm bins), opened once, bodies $\geq 0.5 \text{ mm}^3$; 6 mm next to the support cut; 2 $\times$ -binned maps for beads, full resolution for sand

S2. Void and crack classes in every bead specimen

Figures S1 and S2 give the classes of §2.5.2 of the paper (void, surface crack, body crack inside the bead envelope) at every scan of every bead specimen, in the rendering of Fig. 11 of the paper. Table S2 gives the volumes.

S3. Crack orientation in every bead specimen

The pipeline's body-crack components with their fitted planes, in the rendering of Fig. 12 of the paper, for every loaded scan (Figs. S3 and S4). These renders use the pipeline's per-scan specimen mask, whose boundary artefact the paper describes in §2.5.2; the fitted orientations are read as the principal direction of a damage zone, not as fracture planes.

S4. Ice-path maps of every specimen

Figures S5–S7 give the map of Fig. 18 of the paper for every specimen. For every ice voxel the shortest path through the ice from the support face is computed with a cost inversely proportional to the local ice fraction in

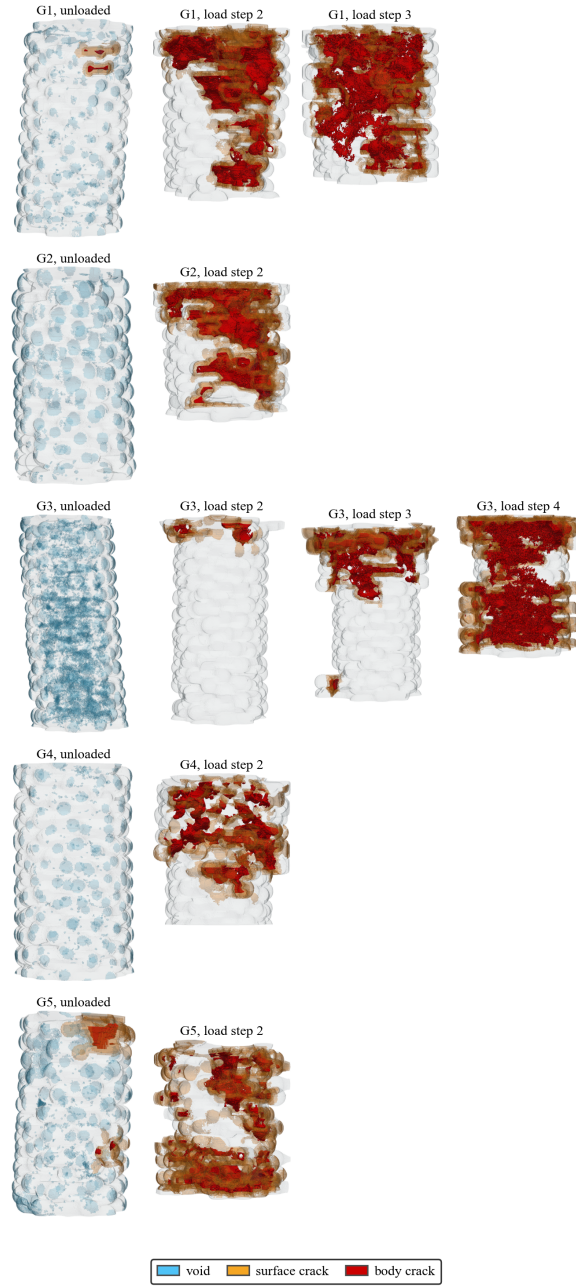


Figure S1: Void and crack classes at every scan of the glass specimens.

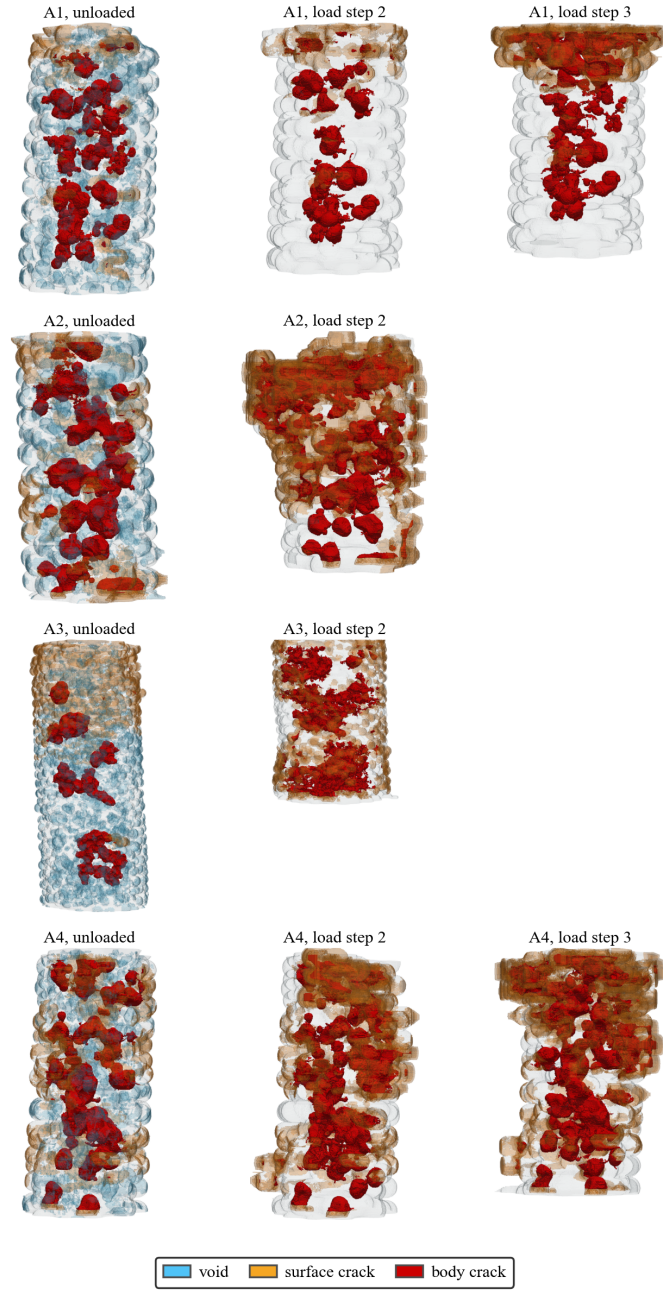


Figure S2: Void and crack classes at every scan of the γ -alumina specimens.

Table S2: Volumes of the classes inside the bead envelope at every scan. The unloaded scan is the baseline of each specimen; the floor of a change is set by A4's first increment (see the paper).

Specimen	scan	envelope (mm ³)	void	surface crack	body crack	separation (mm ³)
G1	1	1356	6.8	3.0	0.1	2.6
G1	2	1358	7.8	61.1	44.9	52.8
G1	3	1383	13.7	49.0	55.9	18.9
G2	1	1148	9.6	0.0	0.0	0.0
G2	2	1195	9.8	52.7	30.4	34.2
G3	1	1514	34.2	0.0	0.0	0.0
G3	2	1486	12.8	6.0	1.4	3.1
G3	3	1614	3.8	49.3	22.9	32.0
G3	4	1344	11.7	92.4	49.3	63.6
G4	1	1242	6.2	0.0	0.0	0.0
G4	2	1220	16.0	30.5	20.2	14.6
G5	1	1146	10.5	7.6	1.0	4.0
G5	2	1180	8.3	80.7	26.0	43.7
A1	1	1358	55.7	12.2	26.8	25.3
A1	2	1309	61.7	9.7	20.3	17.5
A1	3	1314	45.7	33.2	36.8	33.6
A2	1	1388	46.5	21.7	39.8	37.2
A2	2	1477	22.6	103.0	75.5	83.8
A3	1	1834	75.9	18.6	25.2	16.3
A3	2	1279	37.6	37.7	41.2	23.7
A4	1	1447	51.2	28.5	38.4	45.4
A4	2	1500	27.7	72.0	57.9	79.6
A4	3	1528	27.3	102.2	65.0	86.1

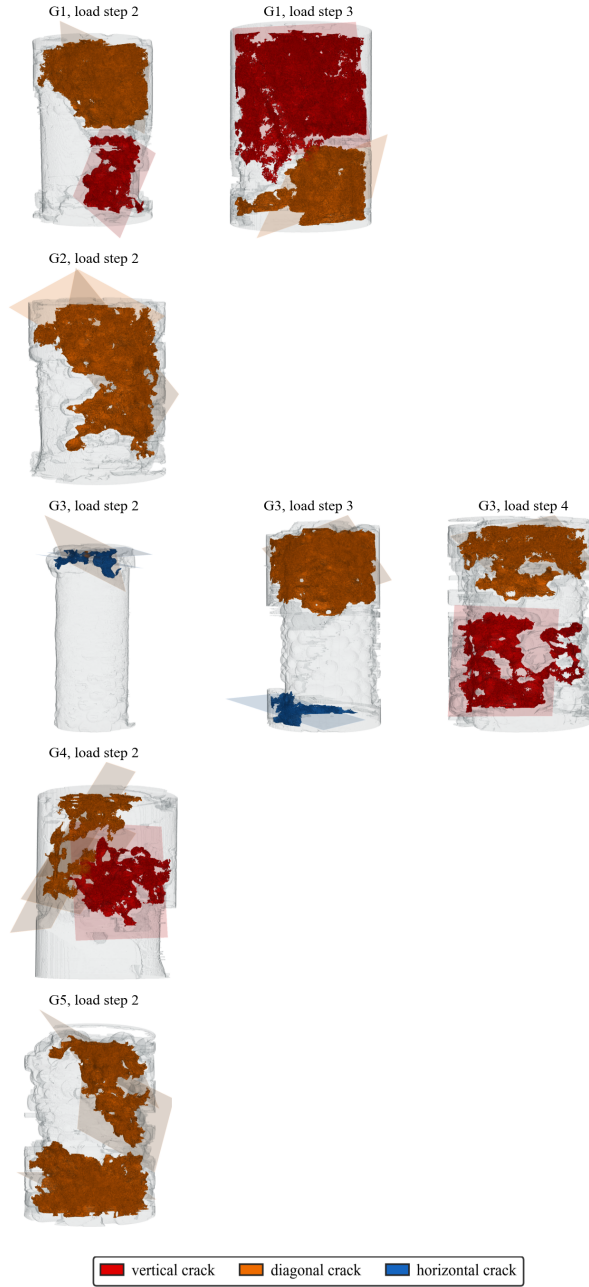


Figure S3: Crack orientation classes of the glass specimens.

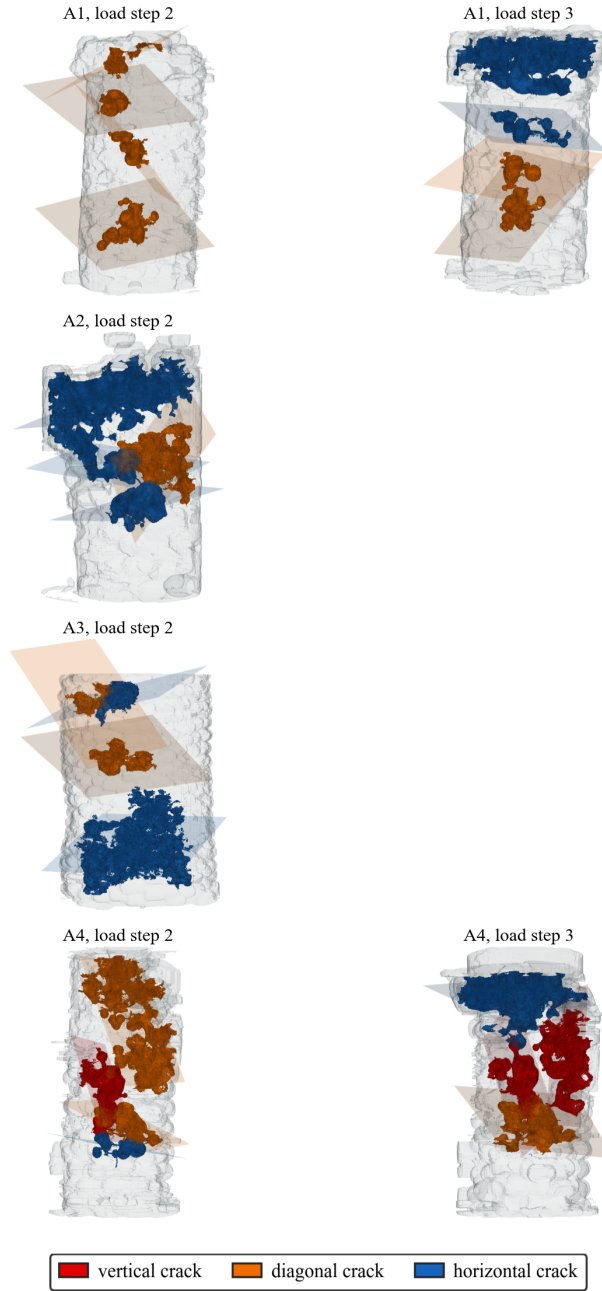


Figure S4: Crack orientation classes of the γ -alumina specimens.

a 0.5 mm window (normalised by the part of the window inside the grain envelope, so the lateral surface does not read as porous). The support face is treated as an equal-potential boundary, as in the diffusion solve: every ice voxel that is the first of its axial line from the support, within 3 mm of it, seeds the path, and the axial distance is measured from that seed surface. The ratio of path to axial distance is divided by its median at the same height in the unloaded column of the same specimen, and ice above the 99.5th percentile of the unloaded column at the same height (bodies of at least 0.5 mm^3) is drawn solid and coloured, the remaining ice as a faint shell, so an unloaded column is blank by construction and a loaded one shows only what exceeds the unloaded range. The map is computed inside the grain envelope on the largest 26-connected ice body, on the $2\times$ -binned phase maps for the beads and the full-resolution phase maps for the sand, whose envelope is pulled in by 1 mm to leave out the ice-rich skin the sand packs against the tube wall; the 6 mm next to the support face are cut, and a line separates the unloaded scan from the loaded ones. The paper shows G1, G3, A4 and S3; the share of the ice above the unloaded range is given in Table S3. In the glass it follows the crack volume of Table S2, G1 and G3 at their later steps far above the rest, G2 and G5 a small share each and G4 nothing. In the γ -alumina A1 and A3 show nothing, and this is not a matter of the threshold, since the distribution of their loaded fields coincides with the unloaded one at every percentile (95th percentile 1.19 against 1.19 in A1, 1.15 against 1.15 in A3), so any threshold that coloured them would colour their unloaded columns as much; the PuMA tortuosity of Table e`ftab:tau` and the crack classes of Table e`ftab:crack` are unchanged in the same two specimens. A2 shows a single body near the support despite the largest crack-volume gain of the four, and A4 the two bead layers under the punch at both steps, present already at its first increment (shortening 0.01), which is the punch bedding into the top layer rather than an interior crack; two thirds or more of the cracked throats of A2 and A4 part at the grain face, so the ice network stays whole. In the sand S1 shows nothing (its second scan is at zero shortening), S2 the outward flow of the head at its third step, and S3 the loaded third of the column with a tongue down the axis.

S5. Per-specimen tables

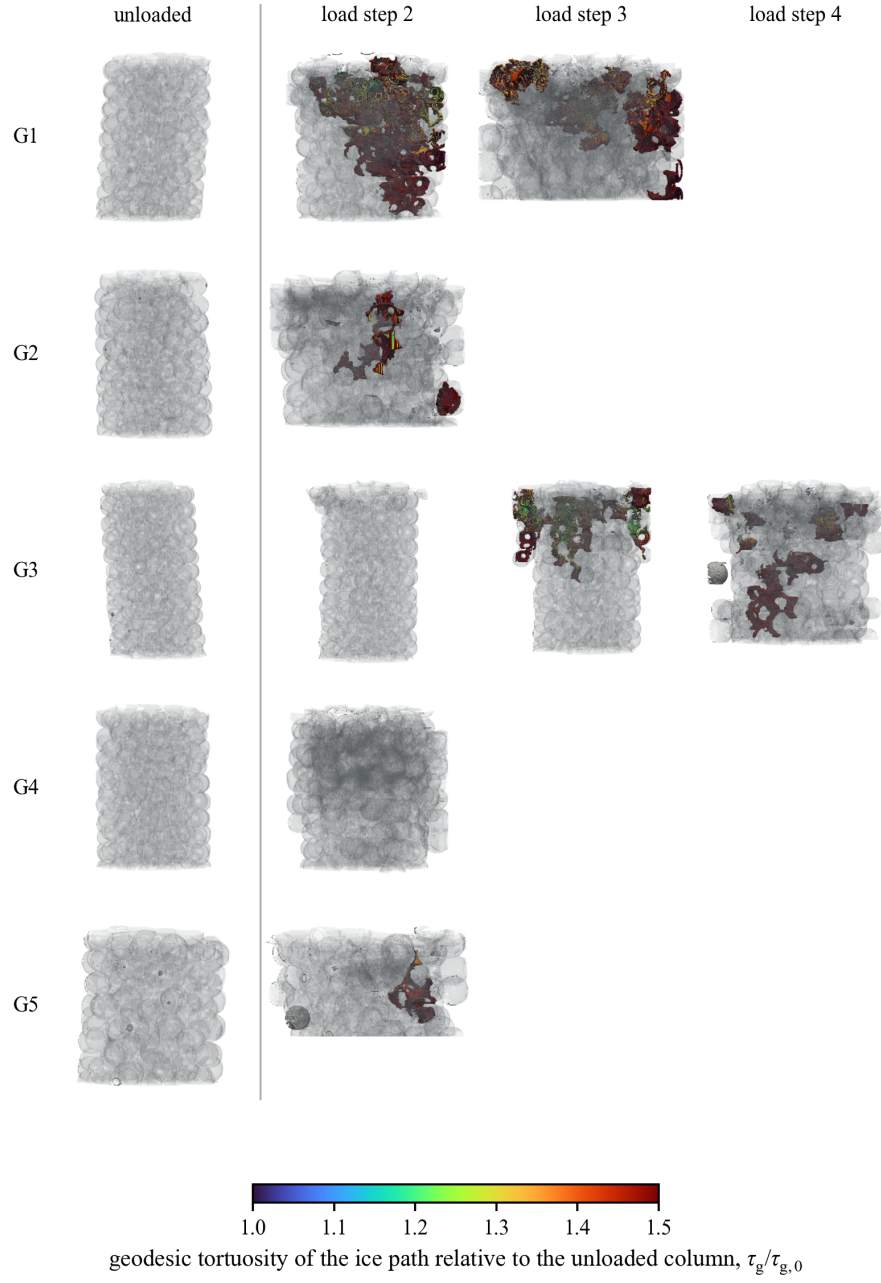


Figure S5: Ice-path maps of the glass specimens.

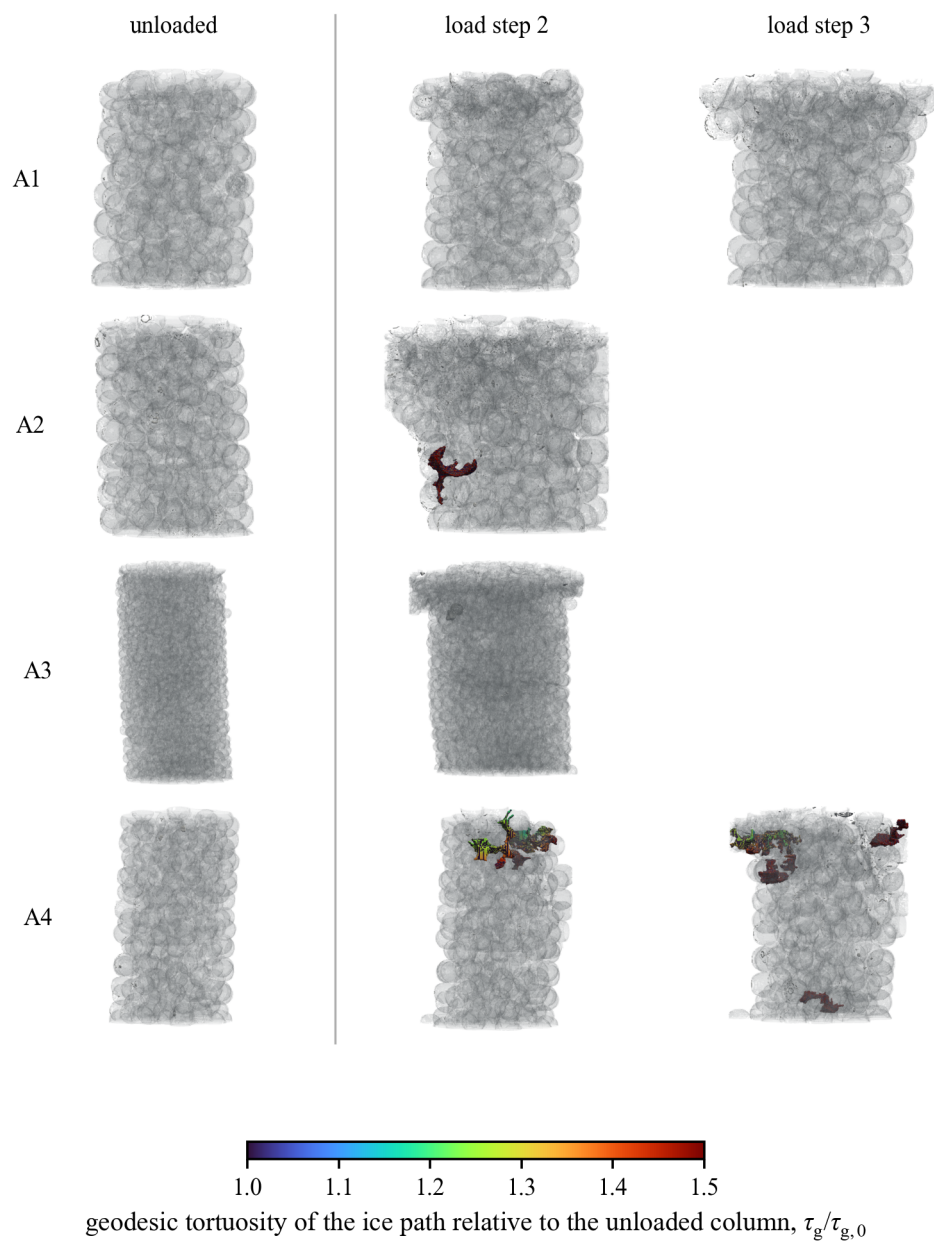


Figure S6: Ice-path maps of the γ -alumina specimens.

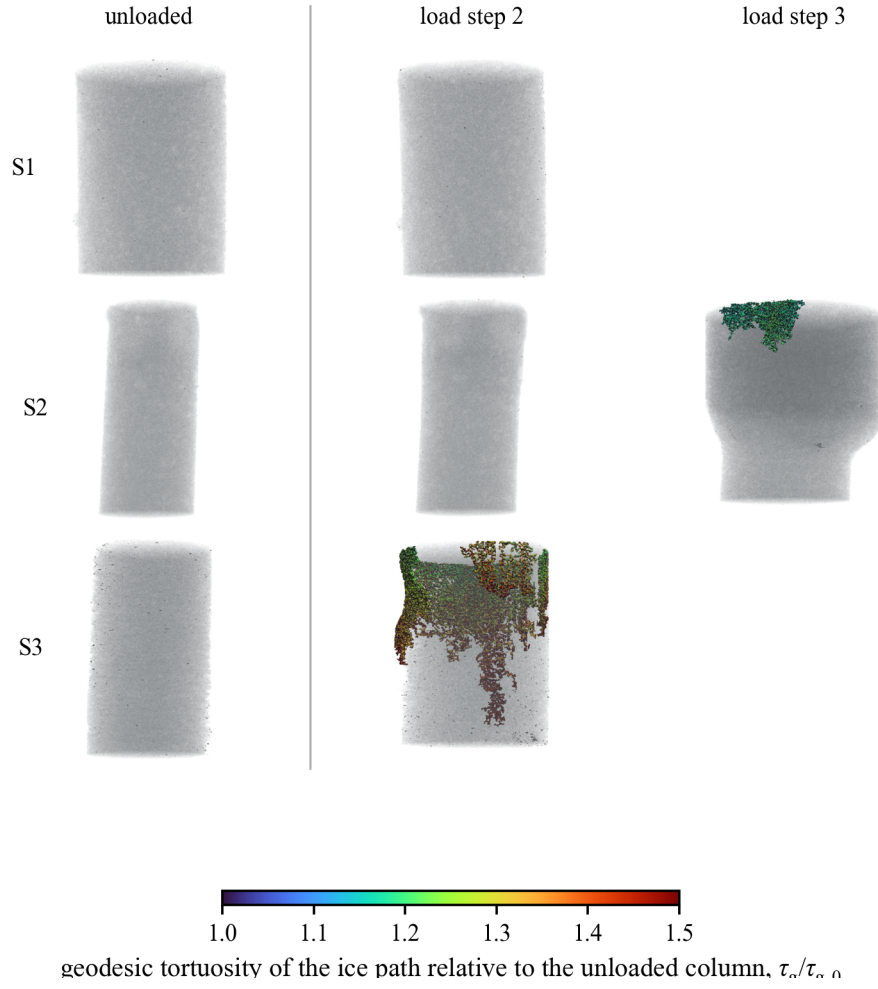


Figure S7: Ice-path maps of the sand specimens.

Table S3: Share of the ice (inside the envelope, above the 6 mm cut) whose relative path exceeds the unloaded column's range at the same height, per scan, in per cent.

Specimen	unloaded	load step 2	load step 3	load step 4
G1	0.00	20.48	7.29	—
G2	0.00	2.44	—	—
G3	0.00	0.00	5.76	5.00
G4	0.00	0.00	—	—
G5	0.00	1.41	—	—
A1	0.00	0.00	0.00	—
A2	0.00	0.26	—	—
A3	0.00	0.00	—	—
A4	0.00	1.58	2.39	—
S1	0.00	0.00	—	—
S2	0.00	0.00	0.35	—
S3	0.00	4.48	—	—

Table S4: Ice tortuosity of every scan from the PuMA continuum diffusion solve, mean and standard deviation over the accepted 128^3 -voxel cubes, with the ice fraction of the cubes and the number accepted of those attempted.

Specimen	scan	shortening	ϕ_{ice} (cubes)	τ_{ice}	sd	cubes
G1	1	0.000	0.399	1.609	0.031	3/3
G1	2	0.161	0.358	1.888	0.000	1/1
G1	3	0.224	0.318	3.737	2.001	2/2
G2	1	0.000	0.412	1.622	0.006	2/2
G2	2	0.206	0.406	1.817	0.050	2/2
G3	1	0.000	0.411	1.610	0.016	3/3
G3	2	0.045	0.416	1.605	0.003	3/3
G3	3	0.102	0.421	1.620	0.027	2/2
G3	4	0.302	0.329	1.840	0.047	3/3
G4	1	0.000	0.469	1.520	0.182	3/3
G4	2	0.219	0.377	1.889	0.198	3/3
G5	1	0.000	0.409	1.608	0.017	2/2
G5	2	0.139	0.359	1.894	0.141	2/2
A1	1	0.000	0.367	1.762	0.032	3/3
A1	2	0.061	0.360	1.818	0.002	2/2
A1	3	0.119	0.368	1.765	0.022	2/2
A2	1	0.000	0.323	1.950	0.088	3/3
A2	2	0.100	0.368	1.949	0.000	1/1
A3	1	0.000	0.368	1.895	0.110	3/3
A3	2	0.367	0.373	2.011	0.039	2/2
A4	1	0.000	0.318	1.920	0.051	3/3
A4	2	0.005	0.334	1.864	0.000	1/1
A4	3	0.077	0.332	2.105	0.151	2/2
S1	1	0.000	0.520	2.330	0.003	2/2
S1	2	0.000	0.528	2.273	0.013	2/2
S2	1	0.000	0.537	2.189	0.108	3/3
S2	2	0.048	0.549	2.168	0.155	3/3
S2	3	0.248	0.536	2.353	0.126	2/2
S3	1	0.000	0.480	2.672	0.114	3/3
S3	2	0.086	0.434	3.235	0.900	3/3

Table S5: Kinematics of every composed field, on the grains carried through the whole chain: median relative displacement with the rigid motion of the column removed, the rigid rotation of the column, the median and 95th percentile of the local rotation, and of the equivalent strain.

Specimen	scans	grains	$ \mathbf{u} - \mathbf{u}_{\text{rigid}} $ (mm)	column rotation ($^{\circ}$)	θ median	θ p95	E_{eq} median	p95
G1	1→2	264/364	1.18	0.8	8.8	15.3	0.315	0.544
G1	1→3	264/364	1.44	1.4	10.0	17.2	0.375	0.695
G2	1→2	302/302	1.65	1.7	9.0	16.6	0.403	0.630
G3	1→2	254/375	0.34	6.3	5.4	17.0	0.006	0.304
G3	1→3	254/375	1.01	8.5	9.1	19.6	0.230	0.555
G3	1→4	254/375	1.35	11.7	10.9	25.5	0.312	0.762
G4	1→2	354/354	1.54	0.9	9.3	19.1	0.323	0.546
G5	1→2	278/278	1.30	1.6	9.1	18.1	0.379	0.573
A1	1→2	223/290	0.70	1.2	10.3	21.5	0.235	0.426
A1	1→3	223/290	1.20	2.0	14.1	26.5	0.370	0.866
A2	1→2	298/298	0.99	1.0	10.1	22.6	0.320	0.700
A3	1→2	2390/2390	0.57	0.2	9.7	36.5	0.271	0.846
A4	1→2	291/330	0.04	2.2	0.4	6.6	0.002	0.190
A4	1→3	291/330	0.97	2.3	10.3	22.6	0.277	0.533

Table S6: Failure-mode census of every bead specimen at each coverage threshold, cohesive/adhesive, with the number of answerable throats, the median gap of those throats and the median share of the throat crack lying against a grain face.

Specimen	answerable	gap median (μm)	face share median	$C_{\text{HI}} = 0.5$	0.7	0.8	0.9
G1	60	208	0.19	53/7	30/30	16/44	8/52
G2	14	180	0.14	7/7	4/10	2/12	0/14
G3	30	215	0.17	2/28	0/30	0/30	0/30
G4	12	246	0.08	6/6	4/8	2/10	0/12
G5	28	220	0.37	8/20	2/26	1/27	1/27
A1	27	202	0.19	20/7	15/12	12/15	9/18
A2	50	213	0.30	27/23	19/31	17/33	10/40
A3	178	215	0.42	80/98	20/158	5/173	0/178
A4	32	210	0.34	19/13	12/20	8/24	3/29
all	431	212	0.33	222/209	106/325	63/368	31/400

Table S7: Breakage-event clustering sweep of every bead specimen: at each neighbourhood radius (in grain diameters) the number of DBSCAN clusters, the largest cluster and the smallest RMS out-of-plane distance of a cluster, each against the median of 400 resamplings of the bonds present and the resampling p . The pre-specified radius is 1.0.

Specimen	R_{nb}/d	clusters / null / p	largest / null / p	RMS/ R / null / p
G1	2.00	1 / 1 / 1.000	115 / 115 / 0.725	2.01 / 1.88 / 0.930
G1	1.25	3 / 2 / 0.443	51 / 64 / 0.850	0.35 / 0.96 / 0.220
G1	1.00	4 / 3 / 0.445	40 / 44 / 0.688	0.35 / 0.50 / 0.260
G1	0.80	3 / 4 / 0.853	11 / 16 / 0.890	0.40 / 0.30 / 0.775
G1	0.60	0 / 1 / 1.000	0 / 7 / 1.000	—
G1	1.50	2 / 1 / 0.350	63 / 107 / 0.955	1.58 / 1.74 / 0.347
G2	2.00	1 / 1 / 1.000	115 / 115 / 0.963	2.19 / 2.07 / 0.963
G2	1.50	1 / 1 / 1.000	112 / 109 / 0.115	2.14 / 2.00 / 0.960
G2	1.25	2 / 3 / 0.920	81 / 71 / 0.225	0.79 / 0.56 / 0.762
G2	1.00	7 / 4 / 0.050	20 / 29 / 0.892	0.21 / 0.34 / 0.105
G2	0.80	3 / 3 / 0.507	10 / 10 / 0.573	0.11 / 0.28 / 0.008
G2	0.60	0 / 0 / 1.000	0 / 0 / 1.000	—
G3	0.60	0 / 0 / 1.000	0 / 0 / 1.000	—
G3	0.80	1 / 0 / 0.003	12 / 0 / 0.000	0.42 / 0.27 / 1.000
G3	2.00	1 / 1 / 1.000	33 / 40 / 0.938	1.21 / 2.05 / 0.058
G3	1.25	1 / 1 / 0.675	27 / 7 / 0.000	1.17 / 0.53 / 0.996
G3	1.50	1 / 2 / 0.995	28 / 16 / 0.080	1.23 / 0.77 / 0.809
G3	1.00	2 / 0 / 0.003	16 / 0 / 0.000	0.62 / 0.41 / 0.864
G4	2.00	1 / 1 / 1.000	122 / 122 / 0.568	2.43 / 2.41 / 0.562
G4	1.50	1 / 3 / 1.000	114 / 58 / 0.000	2.43 / 0.58 / 1.000
G4	1.25	4 / 4 / 0.700	63 / 52 / 0.037	0.34 / 0.34 / 0.500
G4	0.80	3 / 2 / 0.175	13 / 9 / 0.220	0.29 / 0.35 / 0.277
G4	0.60	1 / 0 / 0.225	7 / 0 / 0.120	0.25 / 0.32 / 0.144
G4	1.00	5 / 3 / 0.205	20 / 38 / 0.922	0.28 / 0.32 / 0.360
G5	1.00	4 / 3 / 0.432	58 / 54 / 0.403	0.29 / 0.36 / 0.310
G5	2.00	1 / 1 / 1.000	118 / 118 / 0.843	1.86 / 2.11 / 0.007
G5	1.50	1 / 2 / 1.000	115 / 105 / 0.113	1.77 / 1.09 / 0.540
G5	1.25	4 / 3 / 0.168	70 / 78 / 0.855	0.29 / 0.49 / 0.135
G5	0.80	5 / 2 / 0.040	15 / 14 / 0.472	0.22 / 0.38 / 0.093
G5	0.60	0 / 0 / 1.000	0 / 0 / 1.000	—
A1	0.80	0 / 1 / 1.000	0 / 6 / 1.000	—
A1	1.00	5 / 4 / 0.263	14 / 22 / 0.858	0.30 / 0.36 / 0.307
A1	1.25	6 / 2 / 0.005	22 / 66 / 1.000	0.38 / 0.99 / 0.193
A1	1.50	1 / 1 / 1.000	109 / 105 / 0.065	2.48 / 2.09 / 1.000
A1	2.00	1 / 1 / 1.000	111 / 111 / 0.802	2.46 / 2.21 / 0.998
A1	0.60	0 / 0 / 1.000	0 / 0 / 1.000	—
A2	1.00	9 / 3 / 0.000	20 / 45 / 0.995	0.21 / 0.39 / 0.155
A2	0.60	0 / 1 / 1.000	0 / 6 / 1.000	—
A2	0.80	2 / 2 / 0.750	15 / 11 / 0.242	0.37 / 0.33 / 0.606
A2	2.00	1 / 1 / 1.000	121 / 121 / 0.532	2.67 / 2.29 / 1.000
A2	1.25	5 / 1 / 0.007	67 / 96 / 0.897	0.21 / 1.67 / 0.065
A2	1.50	2 / 1 / 0.058	88 / 114 / 0.995	0.93 / 2.13 / 0.052
A3	2.00	1 / 1 / 1.000	2052 / 2052 / 0.517	4.91 / 4.91 / 0.542
A3	0.60	51 / 33 / 0.000	22 / 19 / 0.320	0.25 / 0.22 / 0.830
A3	1.25	1 / 1 / 1.000	2032 / 2033 / 0.647	4.88 / 4.87 / 0.535
A3	1.00	6 / 5 / 0.438	1849 / 1876 / 0.780	0.07 / 0.28 / 0.015
A3	0.80	70 / 68 / 0.425	139 / 96 / 0.182	0.22 / 0.17 / 0.843
A3	1.50	1 / 1 / 1.000	2042 / 2048 / 0.975	4.89 / 4.90 / 0.427
A4	2.00	1 / 0 / 0.147	7 / 0 / 0.087	0.44 / 0.78 / 0.034
A4	0.60	0 / 0 / 1.000	0 / 0 / 1.000	—
A4	0.80	0 / 0 / 1.000	0 / 0 / 1.000	—
A4	1.00	0 / 0 / 1.000	0 / 0 / 1.000	—
A4	1.25	0 / 0 / 1.000	0 / 0 / 1.000	—
A4	1.50	0 / 0 / 1.000	0 / 0 / 1.000	—